

#Model Comparison & Best Model Selection

##Project Overview

This project is part of the AI & ML Internship Task. The objective is to train multiple Machine Learning models on the same dataset, compare their performance using evaluation metrics, and select the best model.

##Objective

- Train different ML algorithms
- Compare performance metrics
- Detect overfitting
- Select best model
- Save final model

##Dataset Used

Dataset: Titanic Dataset

##Tools & Technologies

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Joblib

##Project Workflow

1.Data Preprocessing

- Loaded dataset
- Handled missing values
- Converted categorical data to numerical
- Dropped unnecessary columns

2. Train-Test Split

- Split dataset into training and testing sets (80/20)

3. Models Trained

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

4. Model Evaluation Metrics

- Accuracy
- Precision
- Recall

- F1-Score

5. Model Comparison

- Created comparison table using Pandas
- Plotted bar chart for performance visualization

6. Overfitting Check

- Compared Train vs Test accuracy

7. Best Model Selection

- Selected model with best test performance and generalization

8. Model Saving

- Saved best model using Joblib

##Output Deliverables

- Model comparison table
- Performance bar chart
- Best saved model (.pkl file)

##Key Learnings

- Importance of comparing multiple models
- Evaluation metrics understanding
- Overfitting detection
- Model generalization
- Industry-level model selection process

##Repository Structure

Model Comparison & Best Model Selection

- dataset.csv
- model_comparison.py
- best_model.pkl
- comparison_plot.png
- README.md

##Conclusion

Among all trained models, the best model was selected based on evaluation metrics and generalization performance. This task demonstrates how ML practitioners choose optimal algorithms for real-world problems.